

Fine-mapping across sparse and sparse-plus-infinitesimal architectures

Iteration 004 benchmark — complete analysis, all six strata

fmbenchmark

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Abstract

This report analyses the complete Iteration 004 fine-mapping benchmark: 45 design rows, 11,250 scenarios, 112,500 region-fits per method, 18 methods, in-sample LD throughout. It spans two genetic architectures — a purely sparse model and a sparse-plus-infinitesimal model — crossed with three annotation conditions, giving six strata that are never pooled. Every quantity derived from posterior inclusion probabilities is defined explicitly before use, including its aggregation level, its tie handling and the noise correction applied to the variance decomposition. The principal findings are: (i) the number of causal variants S dominates the variance budget in every stratum; (ii) the *presence* of annotations matters and their *strength* barely does; (iii) the Functional BEATRICE ablation is monotone and highly significant in all four annotated strata and *exactly zero* in both unannotated ones, over 51,250 paired replicates; and (iv) the annotation-aware methods that lead under the sparse model degrade sharply under misspecification, whereas the region-sharing variant does not — `fb_xregion` rises from third to second of eighteen and overtakes the entire PolyFun family.

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1 Design

1.1 The two architectures

Each simulated region contains exactly S causal variants. The arms differ in what else contributes to the phenotype.

Sparse. Only the S causal variants have effects. Effect sizes are drawn $\beta \sim N(0, \sigma_\beta^2)$ and the residual variance is calibrated to hit the target heritability exactly:

$$\sigma_\epsilon^2 = \text{Var}(X\beta) \frac{1 - \phi}{\phi} \implies \text{PVE} = \frac{\text{Var}(X\beta)}{\text{Var}(X\beta) + \sigma_\epsilon^2} = \phi.$$

Sparse plus infinitesimal. A second genetic component is added, drawn on the **non-causal variants only** ($\alpha_{nc} \sim N(0, 1/m_{nc})$, the BEATRICE formulation). The three variance components are then scaled so that

$$\text{Var}(g_{\text{sparse}}) = p_{\text{causal}} \cdot \phi, \quad \text{Var}(g_{\text{inf}}) = (1 - p_{\text{causal}}) \cdot \phi, \quad \sigma_\epsilon^2 = 1 - \phi.$$

So ϕ is total heritability in both arms, and p_{causal} is **the fraction of heritability carried by the sparse component**. At $p_{\text{causal}} = 0.2$, four-fifths of the genetic signal is infinitesimal.

This has a consequence that governs how the second arm must be read. Under sparse-plus-infinitesimal, *every non-causal variant has a genuine non-zero effect*. The ground truth used for every metric in this report is the set of S sparse causal variants only. A method that correctly detects infinitesimal signal is therefore scored as making a false discovery. **All accuracy and calibration numbers in the second arm measure recovery of the sparse component against a background of real but unlabelled signal**, which is the correct question for fine-mapping but is not the same question as the first arm asks. Cross-arm comparisons of absolute FDR are meaningless for that reason; comparisons of *degradation* between arms are the informative quantity, and are the subject of §6.

1.2 The grid

	rows	annotation	enrichment folds
sparse	1	none	—
	4	binary	2.7, 5.4, 8.1, 10.8
	4	continuous	2.7, 5.4, 8.1, 10.8
sparse + infinitesimal	4	none	— ($p_{\text{causal}} \in \{0.2, 0.4, 0.6, 0.8\}$)
	16	binary	4×4 enrichment $\times p_{\text{causal}}$
	16	continuous	4×4
45 rows $\times$ 250 scenarios $\times$ 10 regions = 112,500 fits per method			

Within every row: $S \in \{1, 2, 3, 5, 10\}$, $\phi \in \{0.05, 0.1, 0.2, 0.4, 0.6\}$, 10 iterations, 10 regions of nominal sizes $\{500, 500, 750, 750, 1000, 1000, 1500, 1500, 2000, 2000\}$, $n = 5000$ individuals.

Causal variants are drawn with annotation-proportional probability. $w_j = \exp(A_j^\top \log \gamma)$, normalised, sampled without replacement. `polyfun_oracle` reconstructs exactly this π_j from the stored truth, which is what makes it a genuine ceiling rather than merely a strong method.

Nominal region sizes are approximate at the top of the range. Regions are drawn from real 1000 Genomes loci, and many do not contain 2,000 usable variants. Measured actual counts:

nominal	median actual	ratio	within 5% of nominal
500	499	1.00	100%
750	749	1.00	96%
1000	999	1.00	97%
1500	1483	0.99	62%
2000	1590	0.80	26%

The true swept range is therefore 499–1590 ($3.2\times$), not 500–2000 ($4\times$). Since ω^2 is range-dependent (§9), region size’s already-small importance is measured over a narrower span than its label suggests. The design remains valid: variation in actual size *within* a nominal class is differentiated away by the σ_u^2 estimator (§9.1) and correctly subtracted. Only the label is approximate.

The unit of analysis. A cell is (row, S , ϕ , nominal region size), aggregating 10 iterations $\times$ 2 regions of that size = 20 fits. There are $45 \times 125 = 5,625$ cells per method. Cell means feed every ANOVA; individual fits feed the calibration counts; replicate-level rows feed the paired contrasts.

Stratification is not optional. Nothing is pooled across architecture or annotation type. Beyond the obvious — a method that cannot use annotations is not comparable to one that can — the unannotated arms serve as negative controls, and the two architectures pose different questions (§1).

2 From posterior inclusion probabilities to numbers

Every method returns, for each region, a vector $\text{PIP}_1, \dots, \text{PIP}_p \in [0, 1]$ — one posterior inclusion probability per variant — together with the truth $y_j \in \{0, 1\}$ known from the simulation. All metrics below are functions of that pair. This section defines each one and justifies the estimator, because several of the obvious implementations are wrong in ways that silently change conclusions.

2.1 Average precision (the ranking metric)

Average precision summarises the whole precision–recall curve without committing to a threshold, which matters here because methods’ PIP scales differ by orders of magnitude: any fixed-threshold comparison partly measures calibration rather than ranking.

Sort the variants by PIP descending. Let k index the distinct PIP values (so that all variants sharing a value form one block), tp_k the cumulative number of causals at the end of block k , n_k the cumulative number of variants, $\text{prec}_k = \text{tp}_k/n_k$ and $\text{rec}_k = \text{tp}_k/S$. Then

$$\text{AP} = \sum_k (\text{rec}_k - \text{rec}_{k-1}) \text{prec}_k, \quad \text{rec}_0 = 0. \quad (1)$$

Three implementation points are load-bearing.

It is a step integral, not a trapezoid. Equation (1) evaluates precision at the operating point reached, matching `sklearn.metrics.average_precision_score`. A trapezoidal rule interpolates between operating points that the classifier never actually occupies and is optimistically biased.

Ties are resolved by block, not by index. Several methods emit many exactly-tied PIPs. Implementing AP as “sort, then read off precision at the causal’s index” lets the sort’s tie-break — here genomic position — decide the score. Because causal variants are drawn uniformly at random this would not introduce a constant bias but would inject pure noise into precisely the comparisons the analysis exists to make. Collapsing tied variants into one block gives every member the same operating point, which is the only tie-handling that is invariant to input order.

AP is computed per fit and then averaged, never on a merged ranking. Pooling all variants from all regions into one ranking and computing a single AP is a different and incorrect quantity: it rewards a method for getting the *relative calibration across regions* right, which is not what region-level fine-mapping is asked to do. The two disagree materially. For two fits with $\text{AP} = 1$ and $\text{AP} = 1/3$, the correct answer is the mean $2/3 \approx 0.667$, whereas the merged ranking of the same data gives 0.700. Both values are checked against hand computation in the test suite.

A fit with no causal variants returns NA rather than 0, and NAs are excluded from means with the count reported alongside, so a method is never credited or penalised for a fit that posed no question.

2.2 Thresholded discovery: FDR and its honest companion

For a threshold t let $\mathcal{S}_t = \{j : \text{PIP}_j \geq t\}$, $n_t = |\mathcal{S}_t|$ and $\text{tp}_t = \sum_{j \in \mathcal{S}_t} y_j$. Then

$$\text{FDR}_t = \frac{n_t - \text{tp}_t}{n_t}, \quad \text{reported jointly with } n_t. \quad (2)$$

Reporting FDR alone is not meaningful: a method that never crosses t has $\text{FDR}_t = 0$ by convention and $n_t = 0$, and would otherwise win the comparison by abstaining. The pair (FDR_t, n_t) is therefore always presented together, and n_t appears in the credible-set analysis (§8) as the follow-up burden.

We also compute the **honesty ratio**. A Bayesian method that emits PIP values is making a falsifiable claim about its own error rate: the expected number of false positives among $\mathcal{S}_t$ is $\widehat{\text{FP}}_t = \sum_{j \in \mathcal{S}_t} (1 - \text{PIP}_j)$. Comparing that to the realised count gives

$$\rho_t = \frac{\text{FP}_t}{\widehat{\text{FP}}_t}, \quad (3)$$

so $\rho_t = 1$ means the method’s own uncertainty statement was accurate, $\rho_t > 1$ means it was overconfident by that factor. This is stricter than FDR: it holds the method to *its own* claim rather than to an external threshold.

2.3 Calibration

Two complementary summaries are used, because each is separately gameable.

Total mass ratio. $\sum_j \text{PIP}_j / S$. A coherent posterior over which variants are causal should place total mass equal to the expected number of causals. A ratio of 2 means the method claims twice as many causal variants as exist.

Banded reliability. Variants are binned by PIP into $[0, 0.1)$, $[0.1, 0.5)$, $[0.5, 0.9)$, $[0.9, 1]$. Within a bin we accumulate the count n_b , the number truly causal c_b , and $\sum \text{PIP}$. Calibration compares the realised rate c_b/n_b to the claimed mean $\sum \text{PIP}/n_b$; this is the reliability diagram in Figure 8. We report ECE_{hi} , the expected calibration error restricted to bins with $\text{PIP} \geq 0.1$, because the $[0, 0.1)$ bin contains the overwhelming majority of variants and an unrestricted ECE is dominated by the trivially-correct statement that almost nothing is causal.

Why both, and why RES. Consider a predictor emitting $\text{PIP}_j = S/p$ for every variant. Its mass ratio is exactly 1 and its calibration error is exactly 0 — it is perfectly calibrated and completely useless. The Murphy decomposition of the Brier score separates this out:

$$\text{BS} = \underbrace{\text{REL}}_{\text{miscalibration}} - \underbrace{\text{RES}}_{\text{resolution}} + \underbrace{\text{UNC}}_{\text{base rate}}, \quad (4)$$

where RES measures how far the method’s conditional rates depart from the base rate. The uniform predictor has $\text{REL} = 0$ and $\text{RES} = 0$. Resolution is therefore the component that detects calibration achieved by saying nothing, and it is reported in Table 3 and Figure 10 alongside REL.

2.4 Credible sets

For a target mass α , variants are added in decreasing PIP order until the cumulative mass first reaches α ; the resulting set size is $|\mathcal{CS}_\alpha|$ and coverage is the indicator that the set contains a causal variant, averaged over fits. When a posterior’s *total* mass never reaches α the set is the whole region and a flag `set_reached` records it — this is a result, not a missing value, and suppressing it would let a maximally diffuse posterior masquerade as a confident small set. Size and coverage are always read together (Figure 11): a set can be made to cover by being large, and small by abandoning coverage.

2.5 The replication structure and the noise floor

The design is a split plot. Two variance components matter: σ_ε^2 , the between-iteration variance at fixed region, and σ_u^2 , the between-region-draw variance. A factor whose levels are compared *within*

the same region draws (namely S and ϕ , which are re-simulated per scenario) sees only σ_ε^2 . A factor whose levels necessarily use *different* region draws — region size, and any term containing it — additionally carries σ_u^2 . Those are “whole-plot” terms.

For a term T with df_T degrees of freedom, the expected sum of squares under the null is $\text{df}_T \lambda_T$ with

$$\lambda_T = \frac{\sigma_\varepsilon^2}{20} + \mathbf{1}\{T \text{ whole-plot}\} \cdot \frac{m \sigma_u^2}{2}, \quad m = 25, \quad (5)$$

the 20 being the fits per cell and the 2 the regions per size class. The noise-corrected share is

$$\omega_T^2 = \frac{\text{SS}_T - \text{df}_T \lambda_T}{\text{SS}_{\text{tot}}}. \quad (6)$$

Why the correction is necessary. Without it, a factor that explains nothing still receives a share proportional to its degrees of freedom, and whole-plot terms receive a large one. In simulation, an uncorrected null factor recorded a 14.6% share where the truth was zero; corrected, it averaged -0.6% over replicates. Negative values are reported as computed and never clamped: a systematically negative share is a diagnostic that $\hat{\lambda}$ is too large, and hiding it would hide the diagnostic.

Estimating σ_u^2 requires a bias correction. The naive pooled estimator over J region draws in K groups is biased low by the factor $c = K(J-1)/(JK-1) = 0.758$ at $J = 4$ — because the pooled mean absorbs part of the between-draw signal. Dividing by c restores unbiasedness; verified over 120 replicates, recovering 0.0438 against a true 0.0400 where the naive estimator gave 0.0332.

2.6 What each “sense” asks

Three different questions are routinely conflated under “which variables matter”. They are kept separate throughout.

Sense V — variance-explaining. Of the variation in a metric across cells, how much does each design factor account for, after the noise correction of (6)? This bounds every claim below it: a noise floor of 50% means no decomposition can explain more than half.

Sense D — decision-relevant. Does changing a factor change *which method you would choose*? Measured as the rate at which the best-performing method flips between adjacent factor levels, split into a raw rate and a gated rate that counts only flips whose margin survives a cross-fitted paired test at $z = 2$. The gap between them is the quantity of interest: it is how much of the apparent structure is noise.

Sense G — diagnostic. For a specific method, which factors hurt *it* relative to what was achievable? Measured by $\kappa = (\text{method} - \text{baseline})/(\text{ceiling} - \text{baseline})$, the fraction of available headroom captured, with SuSiE as baseline and PolyFun-oracle as ceiling.

Sense V and Sense D can disagree entirely, and in this dataset they do. A factor can move a metric a great deal while never changing the winner (every method degrades together), and a factor with a small main effect can flip the winner if it acts differently on different methods. Reporting only one of them overstates what is known.

3 Completeness

Every one of the 18 analysed methods has all 5,625 cells populated, with per-method declared-failure rates of 0.0% and silent all-NA rates of 0.0% across both arms. Three absences are structural and are *not* defects:

- **Funmap** requires annotations and cannot run on either unannotated arm. Its row there is *empty*, not *poor*; no performance claim may be inferred from it.
- **FINEMAP-inf** returns an all-NA posterior without raising an error at $\phi = 0.6$, $S = 1$ in the sparse arm — 45 cells, nine rows $\times$ five region sizes. Recorded via a dedicated `bad_pip` flag so the loss is countable rather than averaged away.
- **CARMA** was dropped mid-iteration on measured cost — 19,000s per scenario against 9,765s for the other eighteen combined, 66% of total runtime — for a differentiator (LD-discrepancy outlier detection) that an all-in-sample design cannot exercise. It is excluded from every analysis here.

A prior failure, and what fixed it. An earlier execution of the sparse-plus-infinitesimal arm recorded 38–50% failed fits for seven Python-backed methods. The cause was environmental, not statistical: the toolchain lived on a filesystem subject to a rolling 30-day purge and decayed mid-run, so activation kept succeeding while imports inside it stopped resolving. Those results were discarded and the arm refitted after moving the toolchain to permanent storage. The run reported here carries a per-task preflight that imports every module the tool sources actually use — derived by AST from the sources rather than listed by hand — and aborts the task if any fails. All 1,800 tasks reported `preflight ok`.

4 Overall accuracy

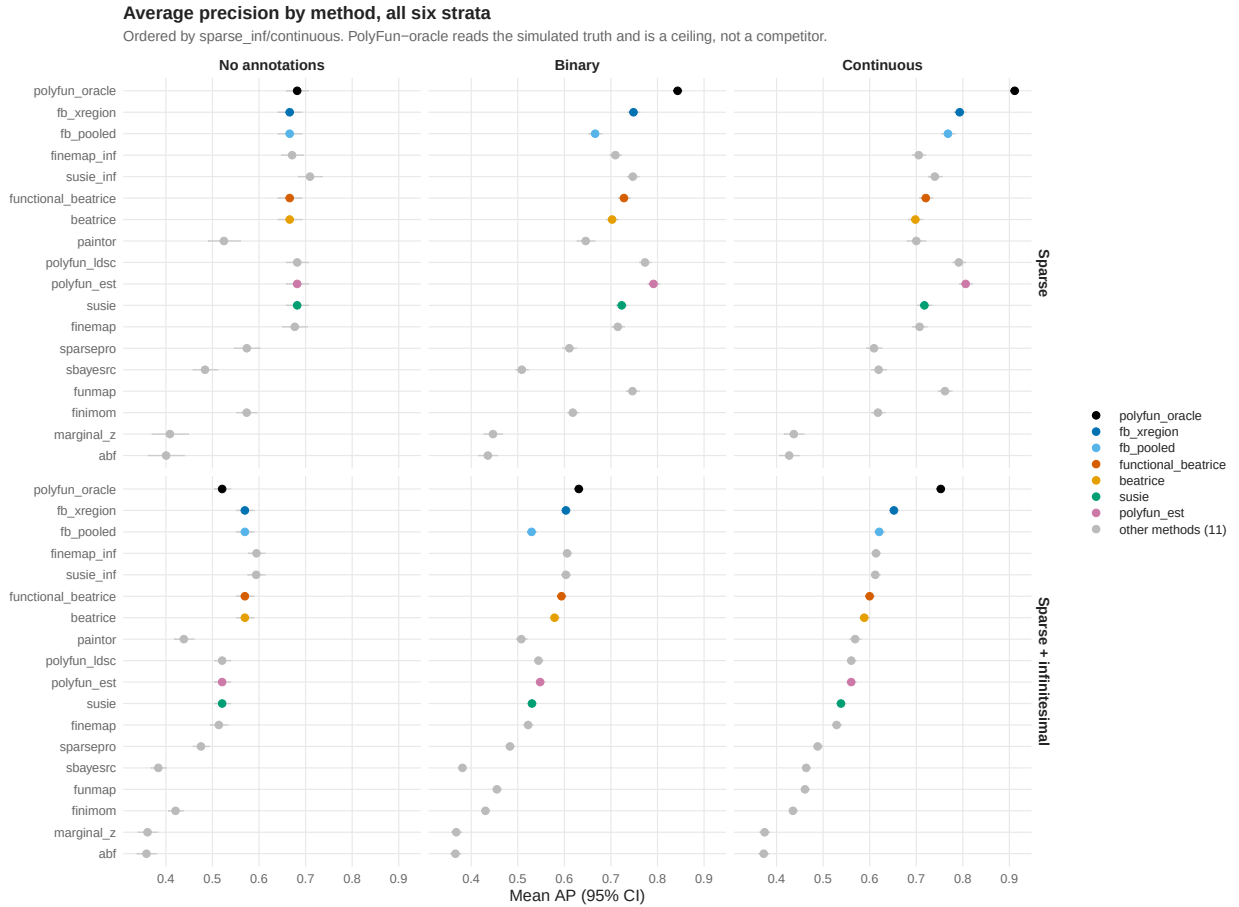


Figure 1: Mean AP with 95% intervals over the 5,625 cells of each method, ordered by the sparse+infinitesimal continuous stratum.

Reading down the two rows of Figure 1, three things change between architectures.

Everything gets harder. Mean AP falls from 0.677 (sparse) to 0.527 (sparse + infinitesimal). This is expected and is not solely a difficulty effect: the infinitesimal background is real signal that the metric counts as noise (§1).

The leaderboard reorders. Under the sparse model the PolyFun family leads among deployable methods. Under misspecification it collapses and `fb_xregion` becomes the strongest non-oracle method on continuous annotations. This is quantified in §6.

FINEMAP-inf comes into its own. It ranks first or second across all three sparse+infinitesimal strata. That is the architecture it explicitly models, and the benchmark rewards it accordingly — a useful internal check that the second arm is doing what it was designed to do.

5 When does each method do well?

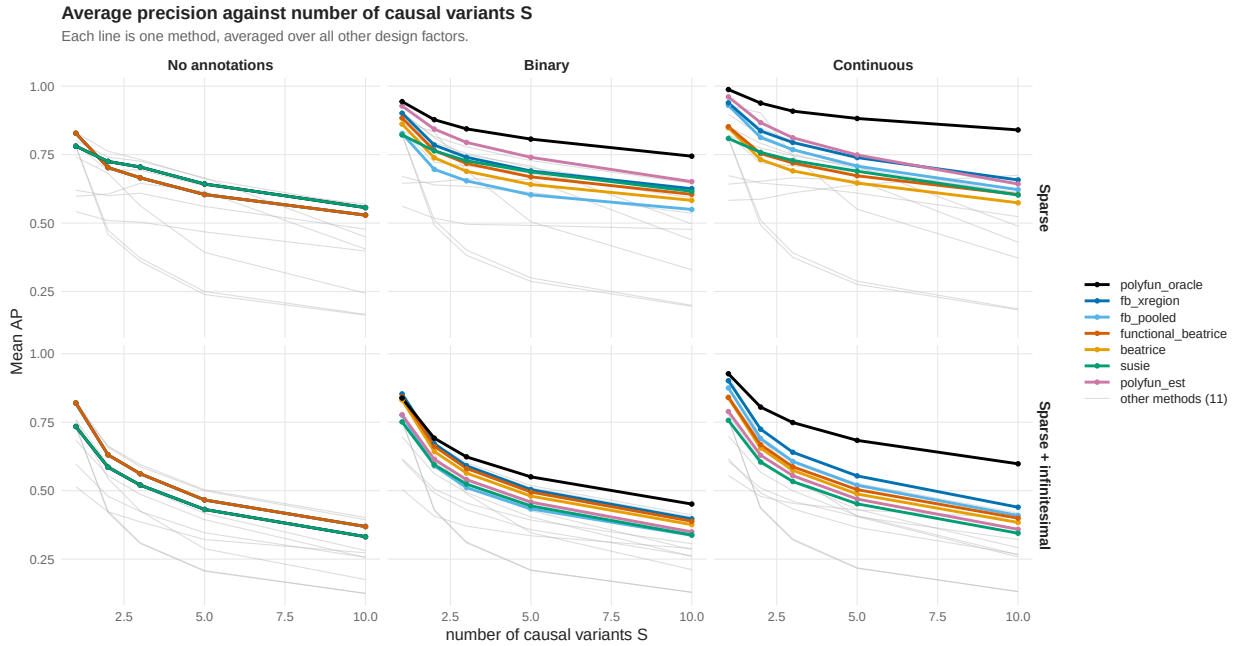


Figure 2: AP against the number of causal variants.

S is the binding constraint in both arms: pooled AP falls $0.780 \rightarrow 0.367$ from $S = 1$ to $S = 10$, a larger swing than any other factor produces. p_{causal} is the second lever in the infinitesimal arm, rising $0.370 \rightarrow 0.640$ from $p_{\text{causal}} = 0.2$ to 0.8 — unsurprising, since it directly controls how much of the signal is the kind the metric scores.

Heritability is not monotone, and that is a real result. Pooled AP rises $0.482 \rightarrow 0.596$ from $\phi = 0.05$ to $\phi = 0.4$ and then *falls* to 0.586 at $\phi = 0.6$. The dip is not an artefact of the FINEMAP-inf gap — it survives excluding that method ($0.5907 \rightarrow 0.5811$) — and it is method-specific. Eight methods degrade at the top of the range, led by `sbayesrc` (-0.088), `sparsepro` (-0.078) and `finimom` (-0.055); ten improve, led by `susie_inf` ($+0.032$). Methods whose priors or variational approximations are tuned for moderate signal do not necessarily benefit from more of it. A benchmark that reported only the pooled mean would have shown a flat top and missed this entirely.

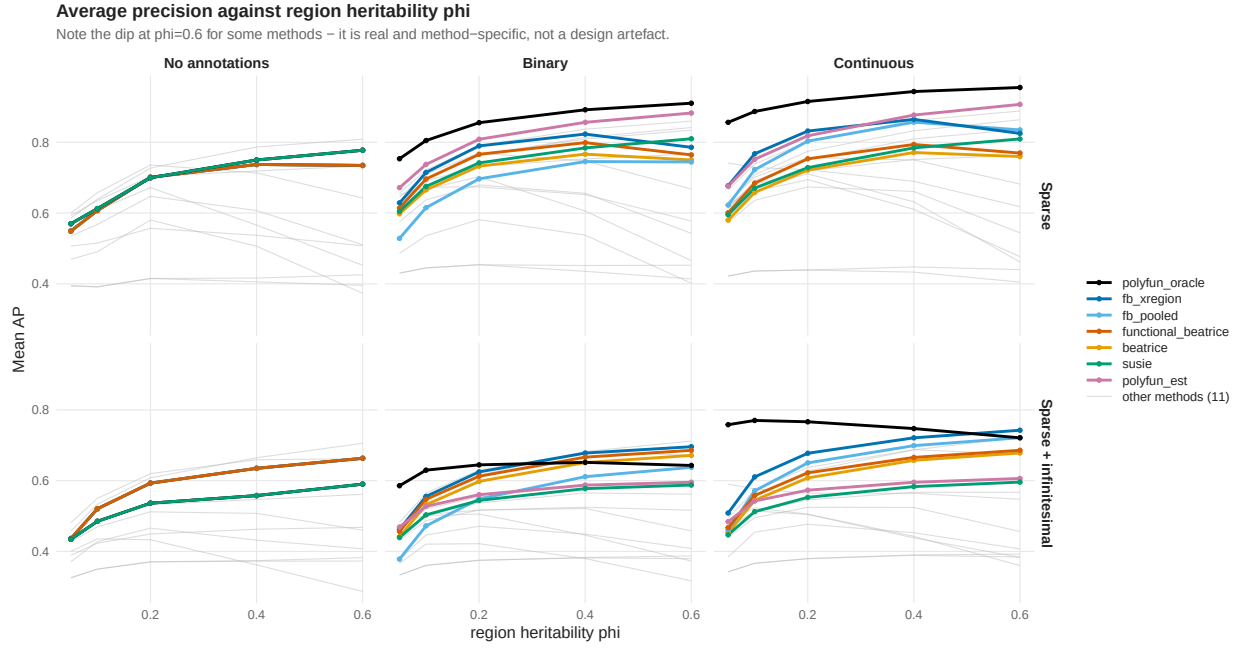


Figure 3: AP against region heritability.

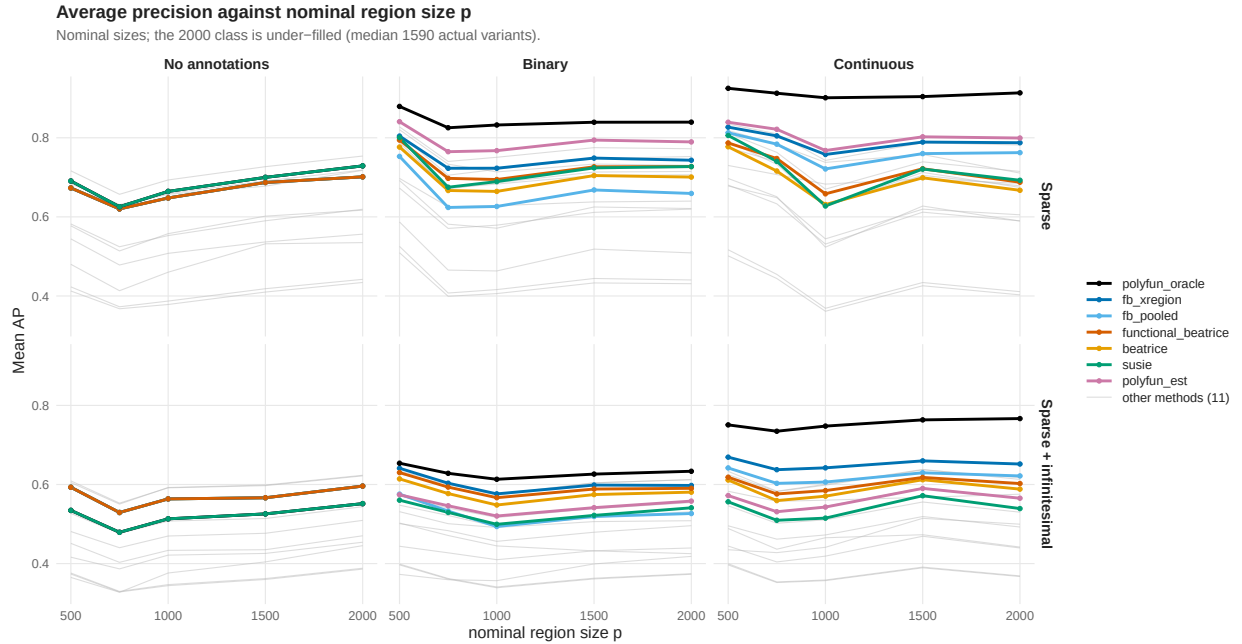


Figure 4: AP against nominal region size. Nearly flat in both arms — over the range actually swept (median 499 to 1590 variants), region size is not what makes fine-mapping hard here.

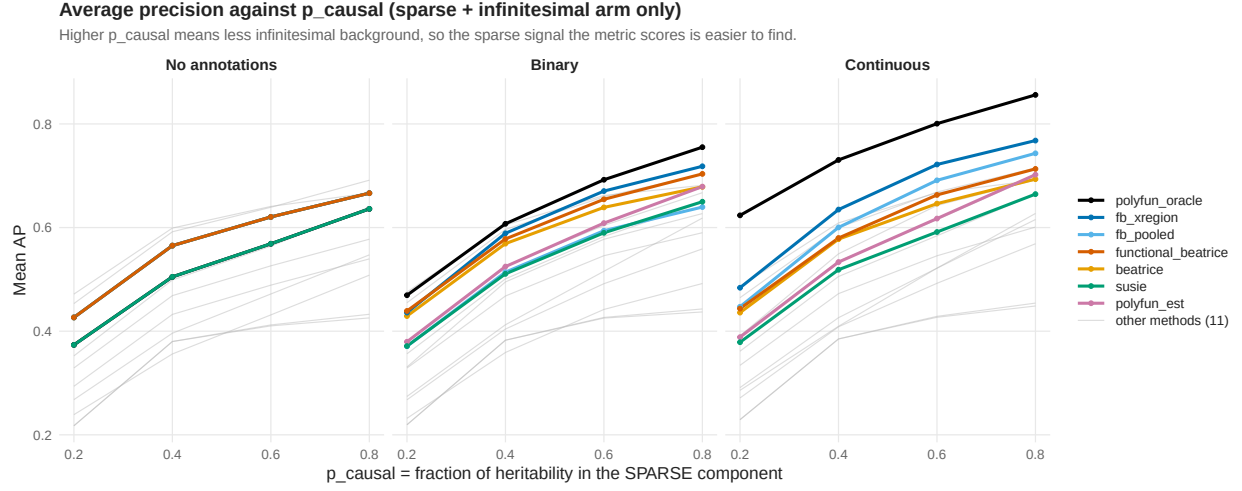


Figure 5: AP against p_{causal} , the fraction of heritability in the sparse component. Sparse+infinitesimal arm only.

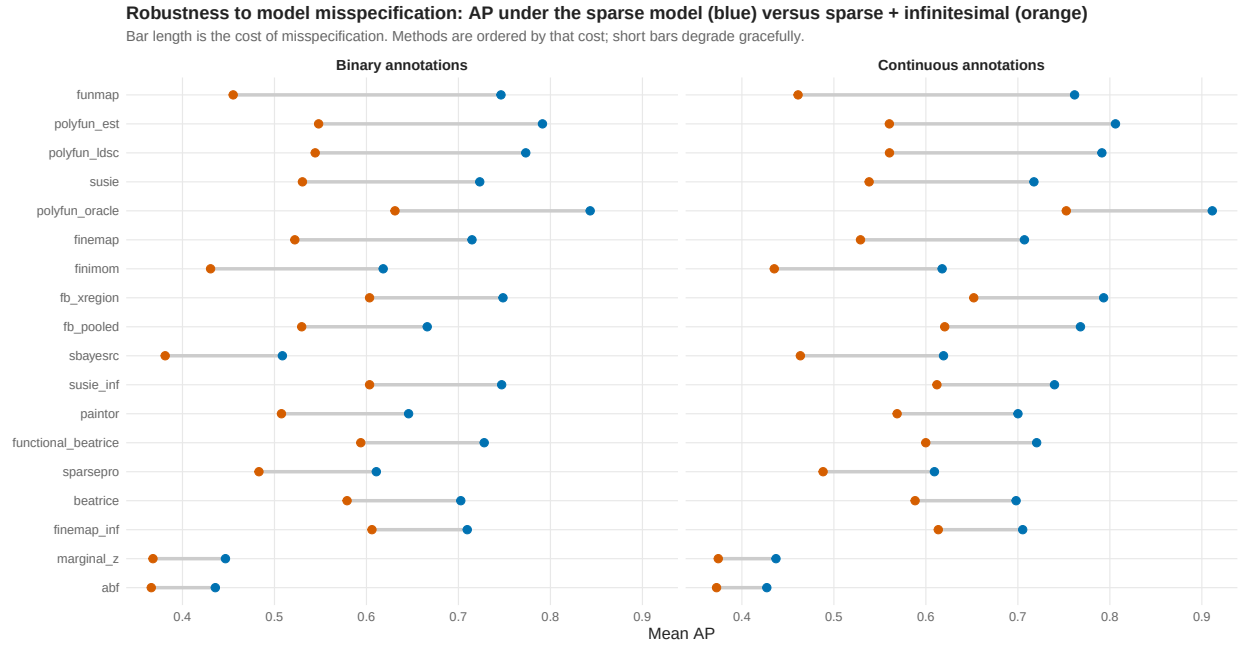


Figure 6: AP under the sparse model (blue) against sparse + infinitesimal (orange), annotated strata. Bar length is the cost of misspecification; methods are ordered by it.

6 Robustness to misspecification

This is the central result of the second arm. Taking continuous annotations:

method	sparse	sparse + inf	
<code>polyfun_est</code>	0.8061 (2nd)	0.5480	FDR@.9 0.005 → 0.35
<code>polyfun_ldsc</code>	0.7914	0.5443	FDR@.9 0.005 → 0.38
<code>fb_xregion</code>	0.7933 (3rd)	0.6521 (2nd)	FDR@.9 0.049 → 0.068
<code>susie_inf</code>	0.7398	0.6120	FDR@.9 0.005 → 0.037

The PolyFun family’s FDR at $\text{PIP} \geq 0.9$ degrades roughly seventy-fold, from about 0.005 to 0.35–0.38, and its ranking collapses. `fb_xregion` loses accuracy too — everything does — but its FDR moves only from 0.049 to 0.068, and it *gains* rank, finishing second of eighteen behind only the oracle.

The mechanism is visible in the design. PolyFun-style methods place a per-SNP prior derived from a model in which annotation-weighted variants are causal and everything else is null. When four-fifths of the heritability sits in unlabelled infinitesimal effects that assumption is badly wrong, and a confident prior built on it produces confident errors. The BEATRICE family’s concrete variational posterior does not commit as hard, and the shared prior head in `fb_xregion` is fitted across ten regions, which appears to regularise it further.

The oracle is itself miscalibrated in this arm. PolyFun-oracle retains the highest AP but its FDR@0.9 reaches 0.36–0.38, because it holds the true *sparse* prior while the infinitesimal component it does not model generates real signal at unlabelled variants. The ceiling is a ceiling for ranking, not for calibration, and κ (§11) inherits that caveat.

7 False discovery and calibration

In the sparse arm the trade-off is the familiar one: the BEATRICE family buys ranking accuracy with roughly $2\times$ mass inflation and an FDR@0.9 an order of magnitude above PolyFun’s, while `susie_inf` and `finemap` stay near mass ratio 1.1–1.2.

Under the infinitesimal model *every* method’s mass ratio rises, to 2.5–4.2. Part of this is genuine overconfidence and part is definitional: mass placed on infinitesimal variants is real posterior belief about real effects, counted here as excess. The two cannot be separated with this ground truth, and the reliability diagrams in Figure 8 should be read with that in mind — the apparent overconfidence of the whole field in the right panel is partly an artefact of scoring against the sparse component alone.

What survives that caveat is the comparison between methods within the arm, since all are scored against the same truth. There, `beatrice` (2.51) and `fb_pooled` (3.11–3.27) are the most conservative of the family and the PolyFun/SuSiE group the least (4.2).

8 Credible sets

Size and coverage are always read together: a set can be made to cover by being large and small by abandoning coverage. Neither number alone identifies which situation a method is in.

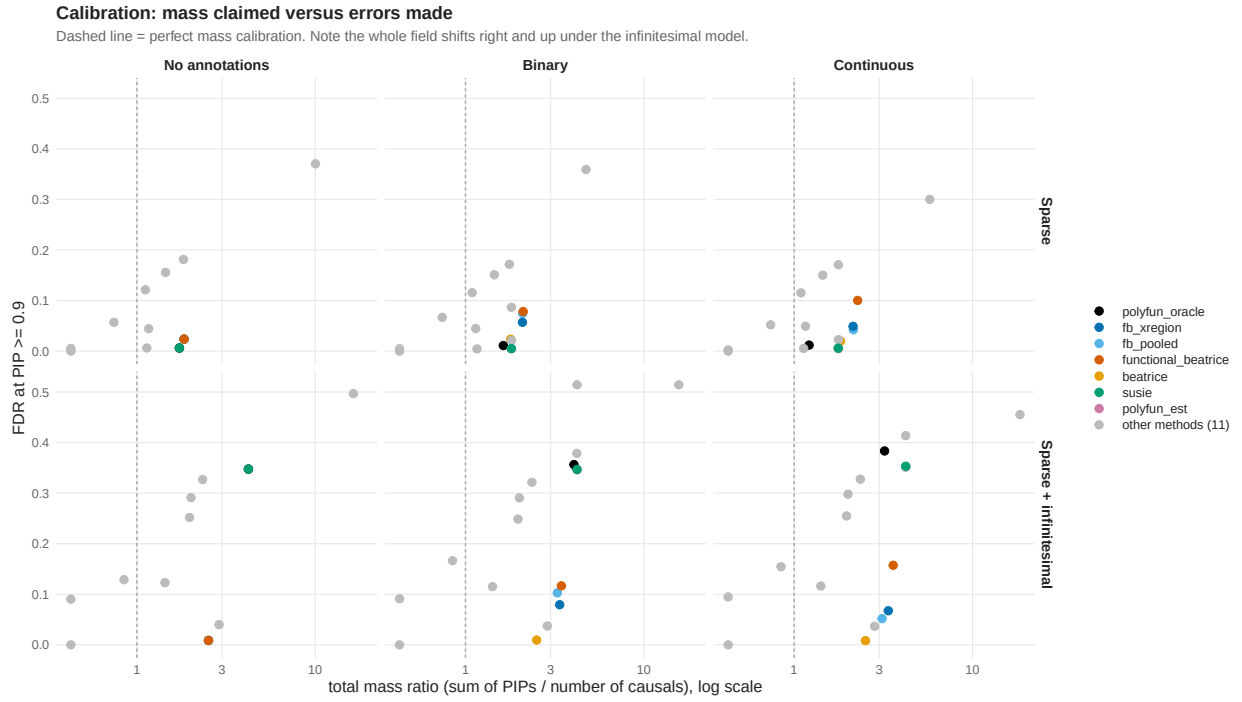


Figure 7: Mass claimed against errors made. The whole field shifts right and up under the infinitesimal model.

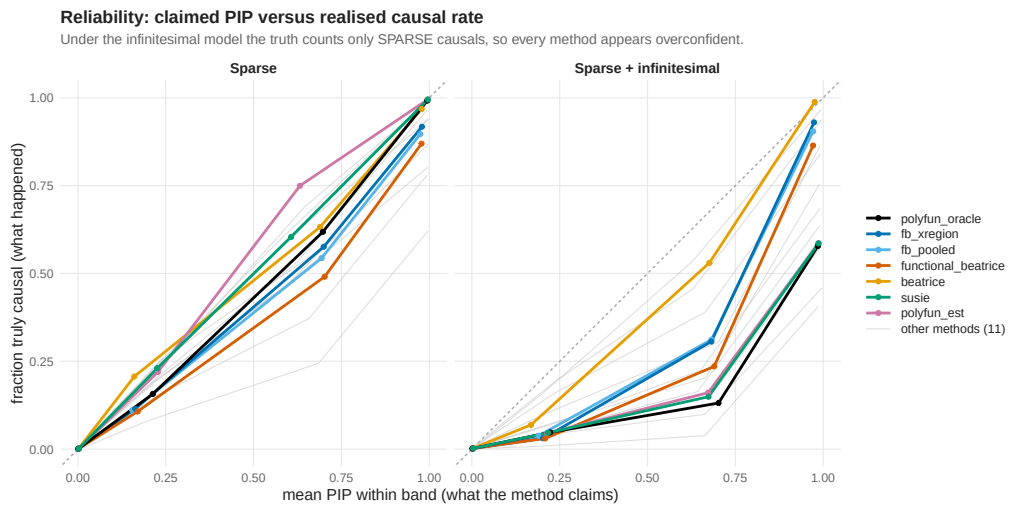


Figure 8: Reliability diagrams, pooled within arm.

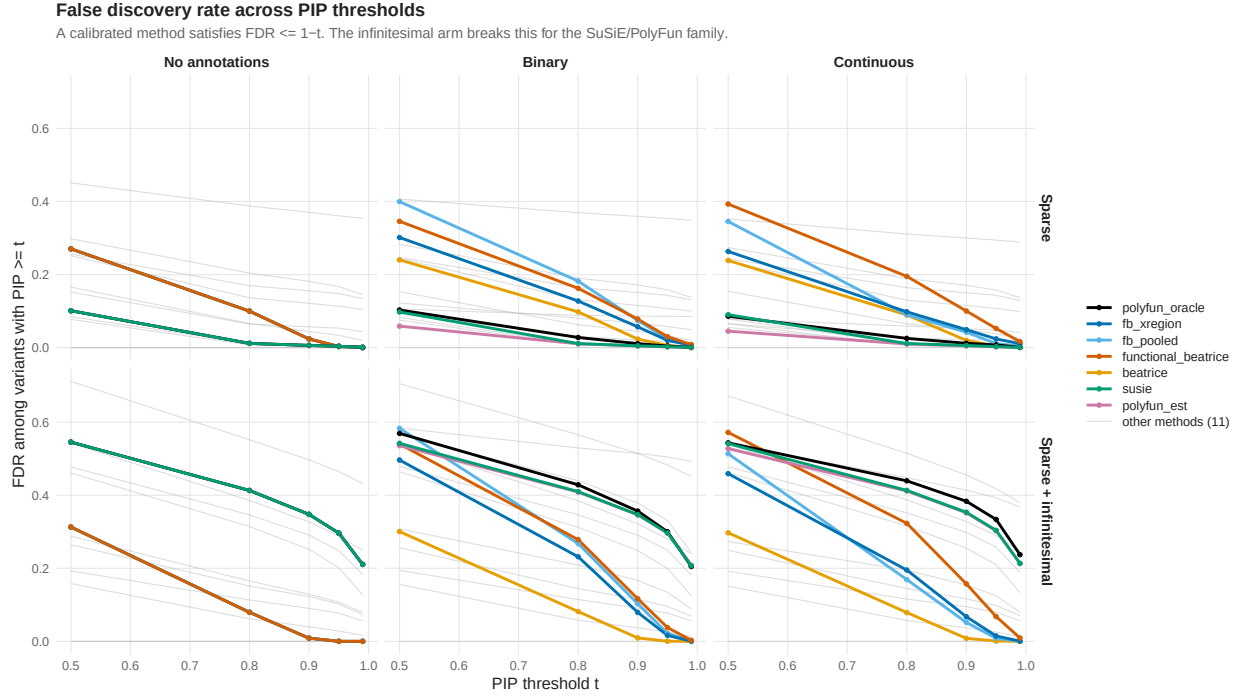


Figure 9: FDR across PIP thresholds. A calibrated method satisfies $FDR_t \leq 1 - t$.

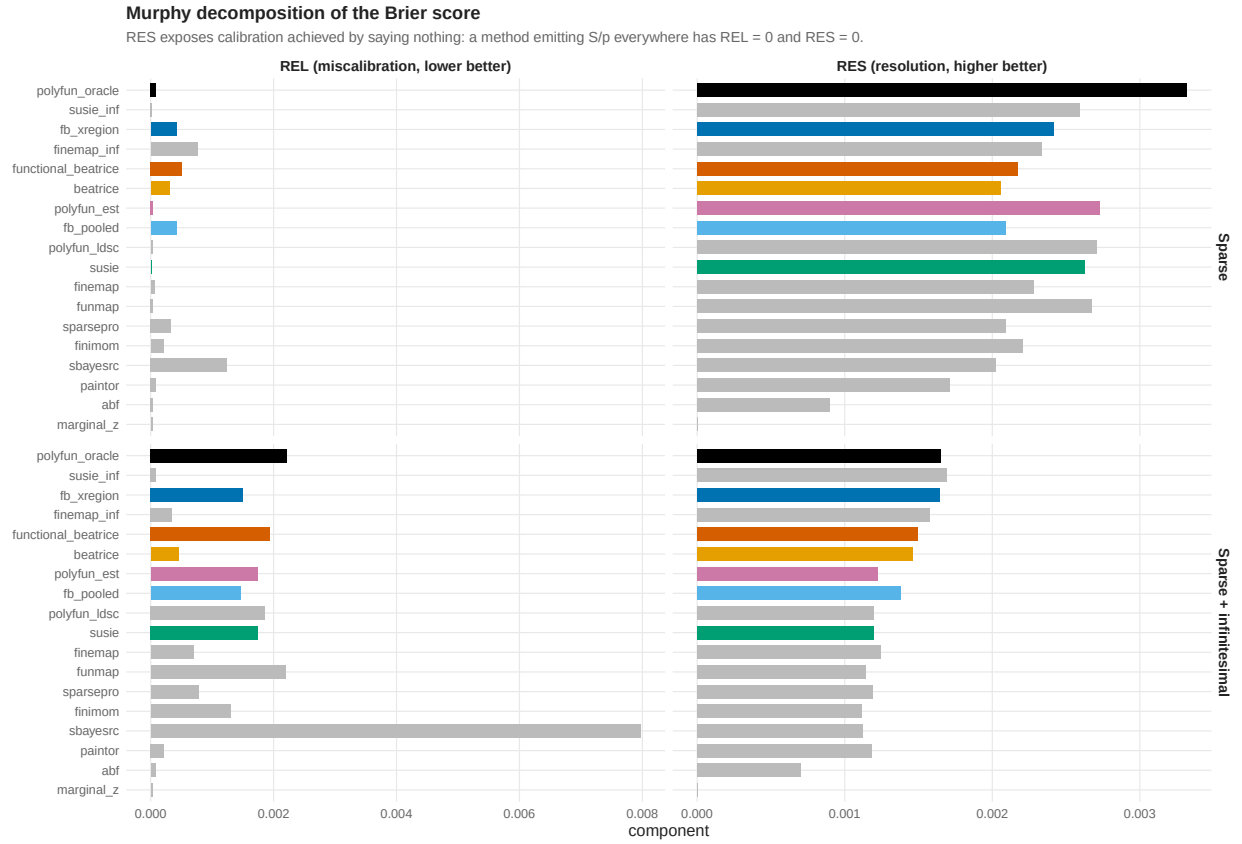


Figure 10: Murphy decomposition. REL is miscalibration (lower better); RES is resolution (higher better). RES is what distinguishes an informative posterior from one calibrated by being vague.

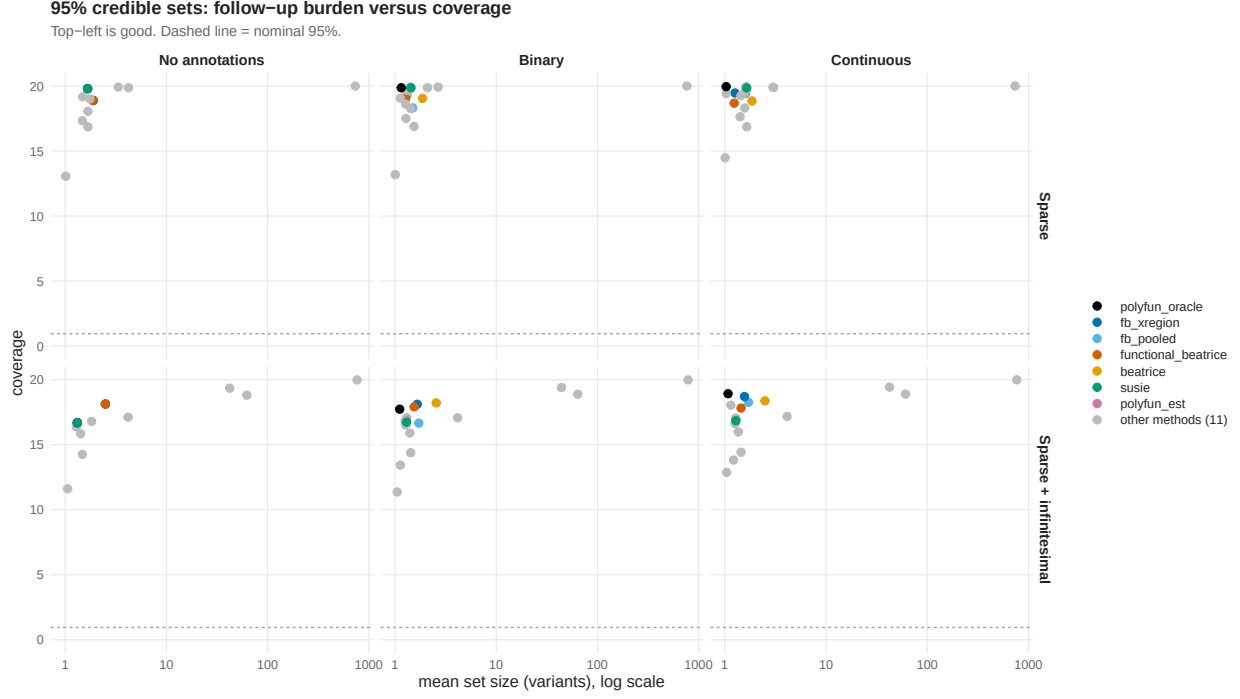


Figure 11: 95% credible sets: follow-up burden against coverage.

9 Sense V: what moves the metric

9.1 How the decomposition is computed, in full

Every number in this section descends from the per-fit AP values of §2.1 by the following chain. Constants are named as they appear in the code: `N_ITER`= 10, `N_REGIONS`= 2 (regions per size class), `N_FITS`= 20, `M_SHARED`= 25, `SUBPLOT_FACTORS`= $\{S, \phi\}$.

Step 1 — form cell means. A cell is the tuple $(\text{job_dir}, S, \phi, \text{region size})$. Its value is the mean of the $N_FITS = 20$ fits it contains: 10 iterations $\times$ 2 regions of that size class. This table (L3) is the only input to the ANOVA; individual fits are used solely to estimate the two variance components below.

Step 2 — estimate σ_ϵ^2 , the iteration variance. Group the fit table by

$$\text{KEY} = (\text{job_dir}, \text{region size}, \text{region_idx}, S, \phi),$$

which pins down *one physical region at one design point*, leaving only the iteration index free. Within each group take the sample variance across its 10 iterations; $\hat{\sigma}_\epsilon^2$ is the mean of those variances. Note what is held fixed: the region, its LD, its variants and its annotations are literally identical across those 10 values, so this is pure replicate error from the phenotype draw.

Step 3 — estimate σ_u^2 , the between-region-draw variance. This cannot be read off directly, because a region draw is never replicated within a size class — there are exactly two, and they differ. It is recovered from their contrast:

1. Average over iterations within each KEY group, giving a region mean $\bar{y}(\text{job}, \text{size}, \text{idx}, S, \phi)$.
2. Reshape wide on `region_idx` and difference the pair: $d = \bar{y}_1 - \bar{y}_2$ for each $(\text{job}, \text{size}, S, \phi)$.

3. In expectation $\text{Var}(d) = 2\sigma_u^2 + 2\sigma_\varepsilon^2/\text{N_ITER}$, so subtracting the second term and halving would give σ_u^2 — except for the bias in Step 4.
4. Do this separately for each region size, then average the five estimates into $\bar{\sigma}_u^2$.

Step 4 — debias the pair variance. The two region draws of a size class are *reused* across all K combinations of (S, ϕ) within a job, so those K differences share a common draw-level offset D_j and are not independent. With J job rows and K cells per job the naive $\text{Var}(d)$ is biased low by

$$c = \frac{K(J-1)}{JK-1}, \quad \hat{\sigma}_D^2 = \max\left(0, \frac{\text{Var}(d) - 2\sigma_\varepsilon^2/\text{N_ITER}}{c}\right), \quad \hat{\sigma}_u^2 = \hat{\sigma}_D^2/2.$$

At $J = 4$ this is $c = 0.758$, i.e. the naive estimator recovers only about three-quarters of the true component. Verified over 120 simulated replicates: corrected 0.0438 against a true 0.0400, where the naive estimator gave 0.0332.

Step 5 — fit the saturated ANOVA on cell means. `aov(response ~ f1 * f2 * ...)` over the stratum's free factors, all interactions included. The design is complete and balanced, so the model is saturated: **there is deliberately no residual line**, every degree of freedom belongs to a named term, and the orthogonal sums of squares partition SS_{tot} exactly. If a **Residuals** row does appear the code warns and drops it, because its presence means the stratum is not balanced and the closed forms below no longer hold.

Step 6 — classify each term.

$$T \text{ is whole-plot} \iff T \text{ contains no factor from } \{S, \phi\}.$$

The logic runs through what a contrast differences away. S and ϕ are re-simulated per scenario *on the same region draw*, so any contrast involving them cancels the draw exactly and sees only iteration noise. A term built solely from region size and enrichment compares values computed on *different* region draws and therefore carries σ_u^2 as well. Note the consequence: **enrich** alone and **enrich:region_size** are whole-plot, whereas **S:region_size** is not, despite containing region size.

Step 7 — the noise expectation and the corrected share.

$$\lambda_T = \frac{\sigma_\varepsilon^2}{\text{N_FITS}} + \mathbf{1}\{T \text{ whole-plot}\} \cdot \frac{\text{M_SHARED} \cdot \bar{\sigma}_u^2}{2}, \quad \omega_T^2 = \frac{\text{SS}_T - \text{df}_T \lambda_T}{\text{SS}_{\text{tot}}}.$$

$\text{N_FITS} = 20$ is the number of fits behind each cell mean; $\text{M_SHARED} = 25 = 5 \times 5$ is the number of (S, ϕ) cells sharing one region draw, and the $/2$ is the two regions per size class. The noise floor is $\sum_T \text{df}_T \lambda_T / \text{SS}_{\text{tot}}$, and it is reported split into its iteration and region parts.

Why cell means rather than one replicate-level ANOVA. Both are defensible in principle; only one is safe here. At replicate level each noise source enters a mean square multiplied by the number of rows sharing that draw — hundreds of them — so subtracting a cell-mean-scale $\sigma_u^2/2$ under-corrects by orders of magnitude *while still looking like a correction has been applied*. Working on cell means keeps every quantity on one scale, which is why the closed form above is exact rather than approximate.

Sobol indices. $V_T = \max(\omega_T^2, 0)$ — this is the *only* place a clamp is applied, and it is applied because the indices are ratios that would otherwise be uninterpretable. Then $S_i = V_i / \sum_T V_T$ and $S_{Ti} = \sum_{T \ni i} V_T / \sum_T V_T$, so $S_{Ti} - S_i$ is the share of variance factor i participates in only through interactions. Everywhere else in this report negative ω^2 is reported as computed.

9.2 The variance budget

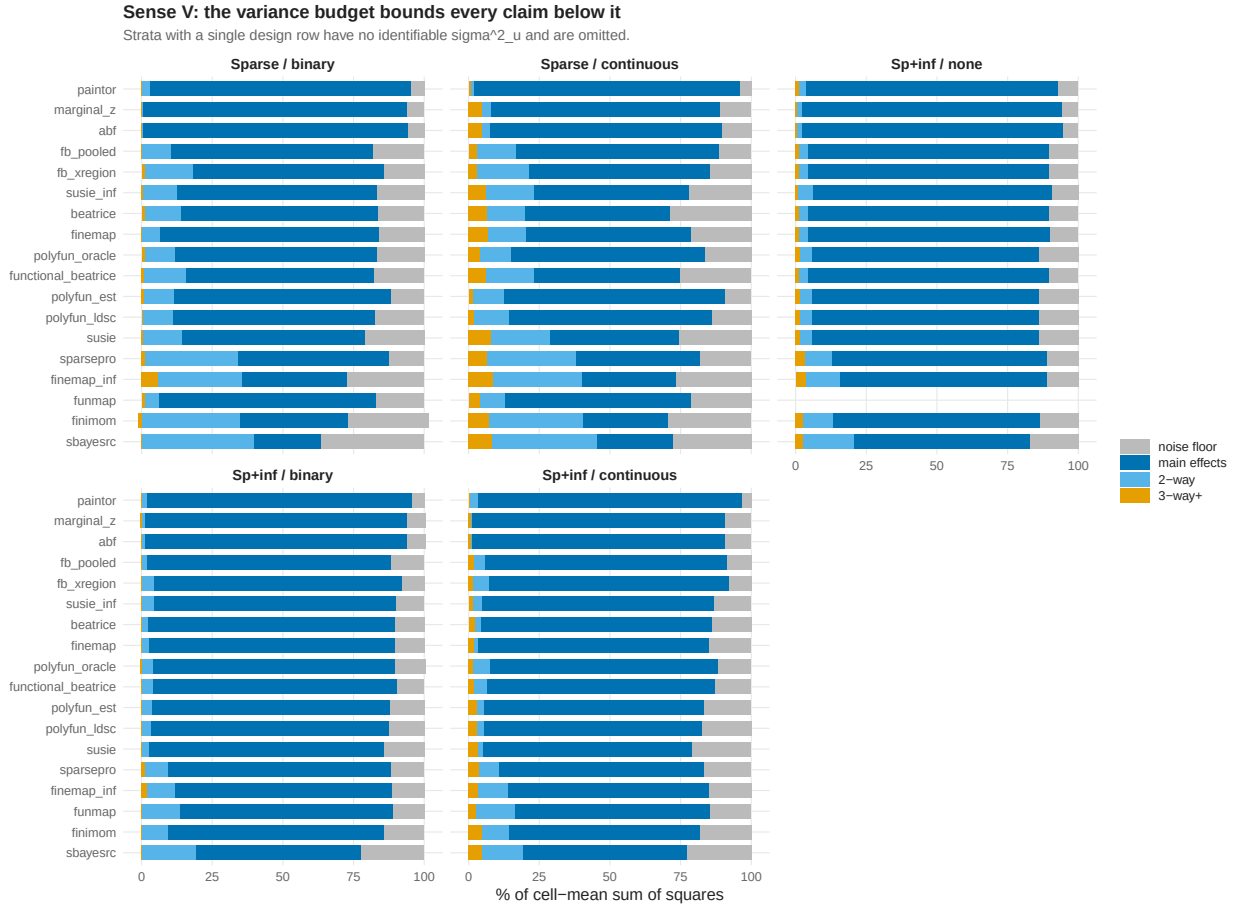


Figure 12: The closed variance budget for AP. Strata whose design contains a single row have no identifiable σ_u^2 and are omitted.

The noise floor runs 17–19% in the strata where it is estimable, with main effects at 60–68% and everything above two-way under 5%. That is the licence for reading the marginal plots of §5 at face value.

Two strata have no budget at all. `sparse/none` contains a single design row, so there is only one group from which to estimate σ_u^2 and the between-region-draw component is unidentifiable. Every corrected share in this section therefore excludes it. Uncorrected shares would be inflated, particularly for whole-plot terms, and are not reported.

9.3 Different metrics have different drivers

Ranking accuracy and mass calibration are both governed by S ; false discovery is governed by ϕ . A benchmark reporting AP alone would call ϕ second-order, when for the metric a practitioner acts on — the error rate among confident calls — it is the leading factor.

Negative shares are reported unclamped. Where $\text{df}_T \lambda_T$ exceeds SS_T the corrected share goes negative, which says the estimated noise for that term is larger than the variation observed. FDR at a fixed threshold is a ratio of small counts and is far noisier per cell than AP, so its decomposition is unreliable at this replication level. Clamping would hide exactly that diagnostic.

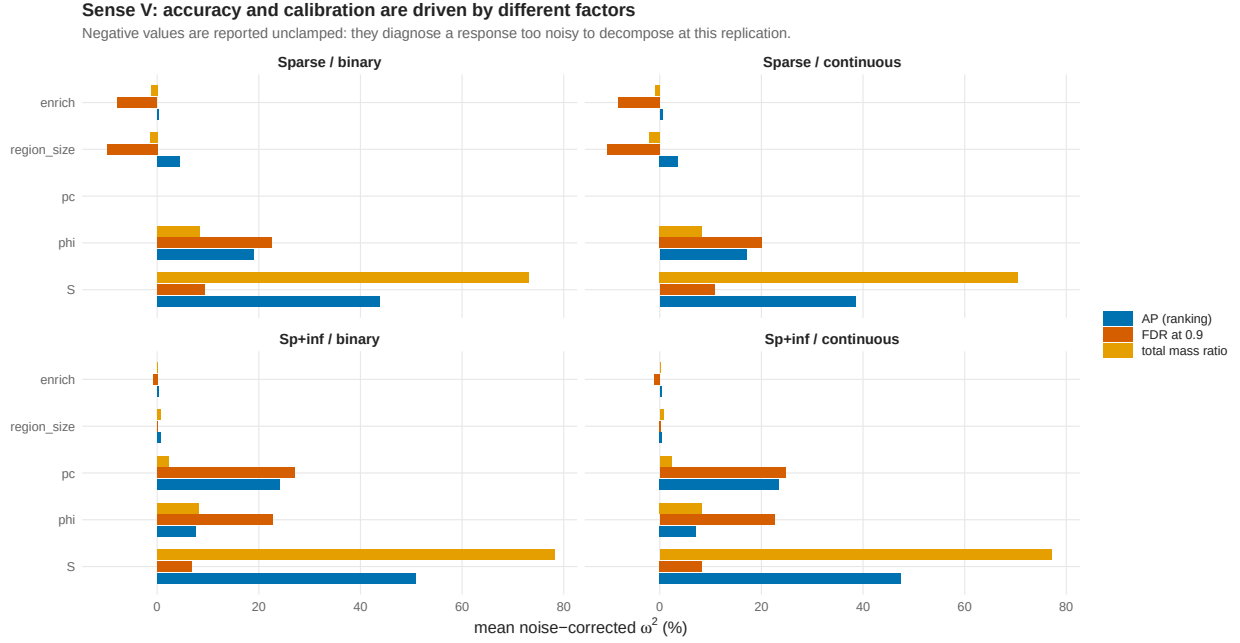


Figure 13: Noise-corrected ω^2 for three responses. The factor ordering changes with the response.

10 Sense D: what changes the decision

10.1 How a cell's winner is decided, in full

The unit is again the cell ($\text{job_dir}, S, \phi, \text{region size}$), but here it is *not* collapsed to a mean. The procedure per cell is:

Step 1 — build the replicate matrix. Reshape to a matrix Y with one row per iteration ($R = 10$) and one column per method, entry $Y_{im} = \text{AP of method } m \text{ on iteration } i$. Because all methods ran on the same simulated data, row i is a genuine paired observation across methods. Columns that are entirely non-finite are dropped; fewer than two surviving columns means the cell is skipped.

Step 2 — the nominal winner. $\text{winner} = \arg \max_m \bar{Y}_{.m}$, the column mean over all 10 iterations. This is the quantity the regime maps display, and on its own it is **not** evidence of anything — the maximum of M noisy means is upward biased by construction.

Step 3 — cross-fitted testing. If $R < 4$ the cell is marked undecided immediately. Otherwise the 10 iterations are randomly permuted (once, under `set.seed(20260806)`, so the result is reproducible) and split into halves A and B . Then, in each direction:

1. Compute column means on the *selection* half.
2. Take the top two methods *by that half only*.
3. On the *other* half form the paired differences $d_i = Y_{i,\text{top1}} - Y_{i,\text{top2}}$.
4. Declare decided iff $|\bar{d}| > z \cdot \text{SE}(\bar{d})$ with $z = 2$ and $\text{SE} = \text{sd}(d)/\sqrt{n}$.

Step 4 — require agreement. The cell counts as decided only if **both** directions (A selects and B tests, then B selects and A tests) declare decided *and* both name the same winner. Any disagreement means the apparent winner depended on which half happened to be used for selection.

Why cross-fitting is necessary. Selecting the top two and testing them on the same data guarantees a positive margin: the selection has already sorted the noise. Splitting selection from testing removes that bias, and requiring both directions to agree removes the residual luck of a single split. This is why the gated rates are so much lower than the raw ones — the raw rate is measuring exactly the bias this procedure is designed to eliminate.

Step 5 — the reported margin. $\bar{d}$ ($\bar{d}$ bar) is computed on *all* 10 rows as $\overline{Y_{\text{winner}} - Y_{\text{runner-up}}}$, using the cross-fitted runner-up. It is a descriptive effect size, not the test statistic.

Step 6 — flip rates. For a factor f : hold every other free factor fixed, walk adjacent levels of f , and record whether the winner changes. `flip_raw` is the fraction of such adjacent pairs whose nominal winner differs; `flip_gated` is the same fraction counting only pairs in which the relevant cells were decided. `pct_cells_decided` is reported separately as the unconditional decidability of the stratum.

10.2 Who actually wins

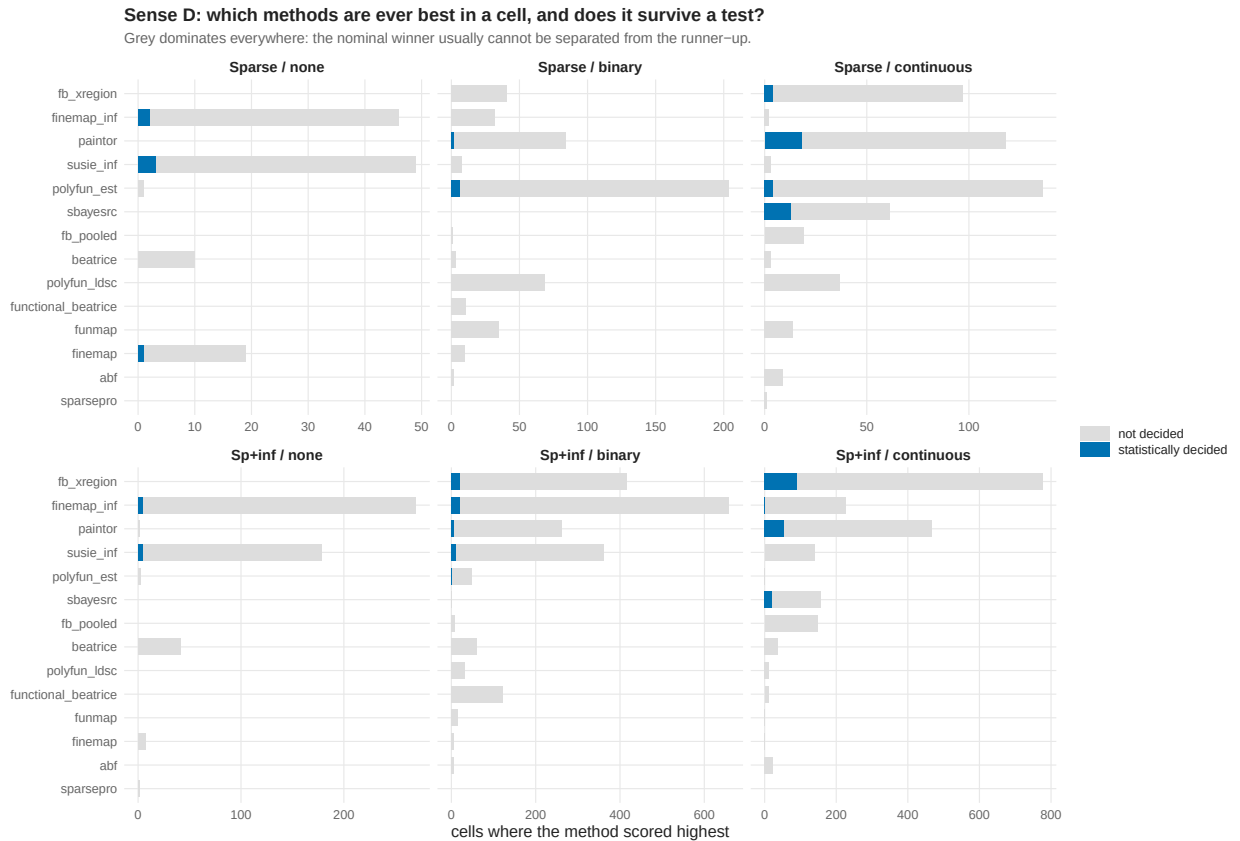


Figure 14: Cells won by each method, split by whether the win survives the cross-fitted paired test.

Grey dominates every panel. The nominal winner of a cell usually cannot be separated from its runner-up, and win counts must not be read as a ranking — most are ties dressed up as wins.

What follows. At 20 fits per cell the benchmark cannot resolve which of eighteen methods is best *within a single cell*, and regime maps built from modal winners are not recommendations. This is a statement about power, not about the methods being equivalent: the paired contrasts of §11 pool tens of thousands of replicates and are decisive. Reporting only one of the two would misrepresent what the benchmark establishes.

11 Sense G: headroom and the ablation

11.1 How κ and the controlled pairs are computed, in full

The input here is the *replicate-level* table (L2): one row per (cell, iteration, method), reshaped so that each (cell, iteration) carries one column per method. Because every method ran on the same simulated data, a row is a set of paired observations — same causal variants, same genotypes, same annotations, same phenotype draw.

Step 1 — headroom. Per row,

$$h = \text{AP}_{\text{polyfun.oracle}} - \text{AP}_{\text{susie}}.$$

`susie` is the annotation-blind backbone — what a method achieves *without* using annotations at all — and `polyfun.oracle` receives the simulator’s true prior weights, so it is what is achievable *with perfect* annotation information. Their difference is the room annotations create on that particular replicate.

Step 2 — the gate. Rows with $h \leq \text{HEADROOM_EPS} = 0.02$ (or non-finite h) are excluded, because κ divides by h and is unstable as $h \rightarrow 0$. The excluded fraction and the median headroom are reported rather than dropped silently (§11.2); on the unannotated arm the gate removes *everything*, which is correct: the ceiling and the baseline are the same method there, so $h \equiv 0$ and κ is undefined by construction.

Step 3 — κ . On surviving rows, for each focus method $m \in \{\text{functional.beatrice}, \text{fb.xregion}, \text{fb.pooled}\}$,

$$\kappa_m = \frac{\text{AP}_m - \text{AP}_{\text{susie}}}{h}.$$

$\kappa = 1$ means m captured all the available headroom; $\kappa = 0$ means it did no better than ignoring the annotations; $\kappa < 0$ means it did worse than the annotation-blind baseline on that replicate.

Step 4 — the controlled pairs, which are NOT gated. Two contrasts are formed on *all* rows, with no headroom gate, because they are differences rather than ratios and have no denominator to destabilise:

$$\Delta_{\text{annot}} = \text{AP}_{\text{functional.beatrice}} - \text{AP}_{\text{beatrice}}, \quad \Delta_{\text{joint}} = \text{AP}_{\text{fb.xregion}} - \text{AP}_{\text{functional.beatrice}}.$$

Each isolates exactly one mechanism, because the two methods being differenced are identical in every other respect: same inference network, same hyperparameters, same data, same seed. Δ_{annot} turns the annotation prior on; Δ_{joint} changes that prior from per-region to shared-across-regions.

Step 5 — inference on the pairs. For a group of n paired rows, $\bar{\Delta}$, $\text{sd}(\Delta)$, $\text{SE} = \text{sd}/\sqrt{n}$ and $t = \bar{\Delta}/\text{SE}$. The pairing is what supplies the power: the between-scenario variance that dominates marginal comparisons cancels exactly, leaving only the within-replicate difference. This is why §11 resolves effects of 0.02 AP decisively while §10 cannot resolve a 0.13 gap between two *different* methods in a single cell — n differs by three orders of magnitude and the nuisance variance has been removed.

Step 6 — conditional breakdowns. The figures in §11.4 are $\bar{\Delta}$ recomputed within levels of one design factor at a time, pooling the two annotated strata. 114 of the 22,500 Δ values are missing (0.5%), all at $S \in \{5, 10\}$, corresponding to the BEATRICE family’s small fit-failure rate; those rows are dropped pairwise.

11.2 Headroom exists in both arms

arm	annotation	median headroom	% ≤ 0	replicate rows
sparse	none	0.0000	100.0%	1,250
sparse	binary	0.0840	25.5%	5,000
sparse	continuous	0.1662	20.6%	5,000
sparse + inf	none	0.0000	100.0%	5,000
sparse + inf	binary	0.0667	21.2%	20,000
sparse + inf	continuous	0.1915	13.9%	20,000

Despite the oracle’s poor calibration under misspecification (§6), its *ranking* advantage over the baseline is intact, so κ remains well defined in the annotated strata. The largest median headroom of any stratum is sparse+infinitesimal / continuous at 0.1915 — there is *more* for annotations to exploit when the model is misspecified, not less. Both unannotated arms have headroom identically zero, so the gate removes them entirely; that is correct, since the ceiling and the baseline are the same method there.

11.3 The ablation

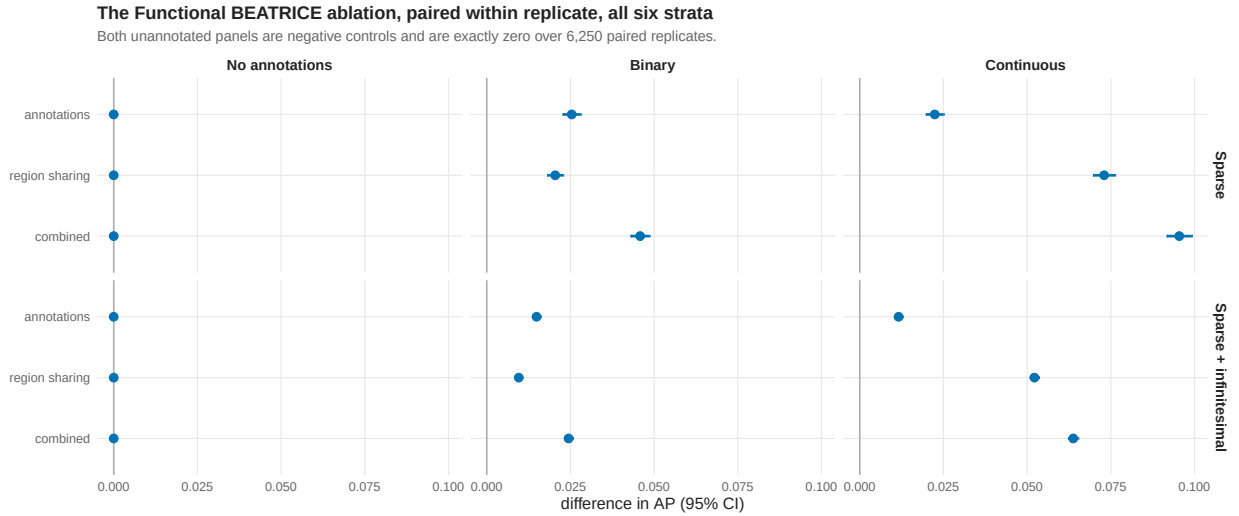


Figure 15: Paired contrasts with 95% intervals, all six strata.

arm	annotation	annotations	region sharing	combined	n
sparse	none	0.0000	0.0000	0.0000	1,250
sparse	binary	+0.0254	+0.0205	+0.0458	4,980
sparse	continuous	+0.0224	+0.0730	+0.0954	4,960
sparse+inf	none	0.0000	0.0000	0.0000	5,000
sparse+inf	binary	+0.0149	+0.0096	+0.0244	20,000
sparse+inf	continuous	+0.0116	+0.0522	+0.0638	20,000

All twelve annotated contrasts are positive with t between 17 and 82 (Table 7). Both unannotated arms are **exactly** 0.0000 ± 0.0000 over 6,250 paired replicates — the three variants are

bit-identical there, so the measured gains cannot be an artefact of the extra machinery, of additional parameters, or of run-to-run variation.

Gains are smaller under the infinitesimal model, which is expected: the background is variance no annotation can explain. But the *ordering* inverts in an informative way — region sharing contributes four times what annotations alone do on continuous annotations there (+0.052 against +0.012), against roughly three times in the sparse arm. **The harder and more misspecified the problem, the more the sharing matters relative to the annotations themselves.**

11.4 Where the gains come from

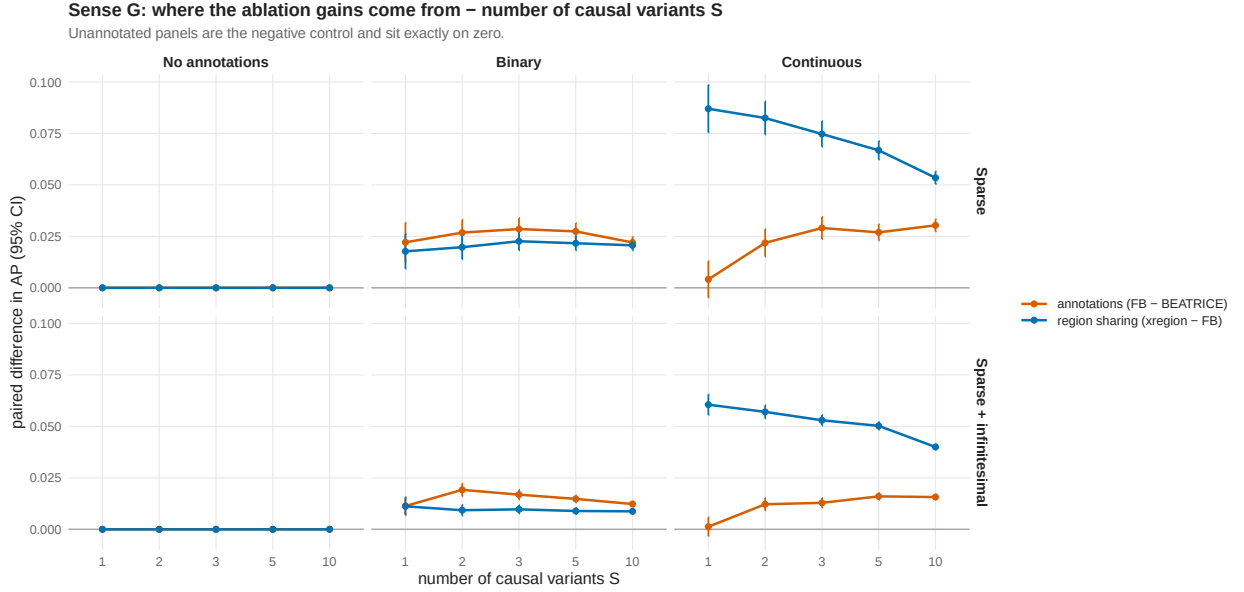


Figure 16: Both contrasts against S .

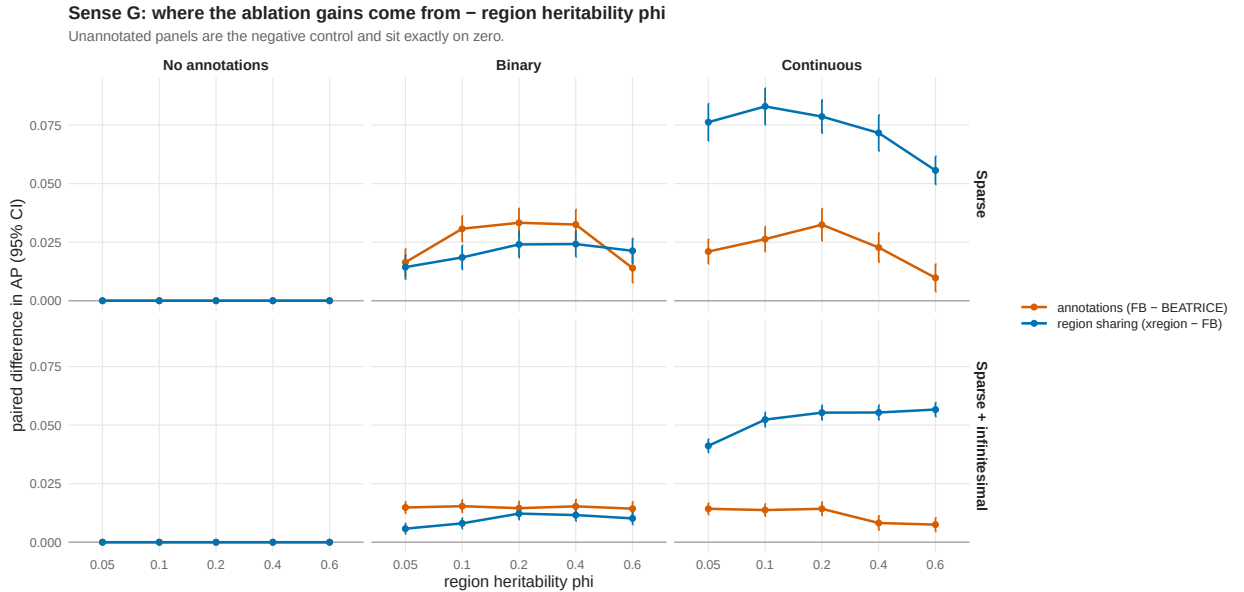


Figure 17: Both contrasts against ϕ .

Region sharing is worth most where regions contain fewest causals and decays as S grows. That is what the mechanism predicts: a region with a single causal variant carries almost no

internal information about which annotations matter, so borrowing the enrichment estimate from nine other regions is worth a great deal, while a region with ten causals can estimate it locally.

11.5 κ is reported but no conclusion rests on it

After gating, κ 's standard deviations exceed unity against means well below it. The denominator is small in many surviving cells and the ratio is correspondingly unstable. Every conclusion in this section rests on the controlled pairs, which are differences rather than ratios.

The κ variance decomposition is unavailable. The replicate-level table does not carry the region index, so σ_u^2 cannot be estimated for the derived κ and Δ responses. The code refuses to report an uncorrected decomposition, which would inflate whole-plot terms by $m\sigma_u^2/2$ per degree of freedom.

12 Validity checks

Mechanically verified on this run; all pass.

1. In-sample LD only — zero rows carrying a reference-panel size.
2. 45 design rows present, matching the grid.
3. Balanced cells per analysed method — exactly 5,625 for all 18.
4. CARMA excluded by design and reported as such.
5. No method entirely absent outside the unannotated arms.
6. `polyfun_est = susie` on both unannotated arms, exactly.
7. `polyfun_ldsc = susie` on both unannotated arms, exactly.

Additional checks performed for this report, beyond the pipeline's own:

8. **Truth consistency:** `n_causal = S` for all 2,031,230 fit rows; zero mismatches.
9. **Grid completeness:** exactly 11,250 distinct $(\text{row}, S, \phi, \text{iter})$ combinations; `region_idx` takes only the values 1 and 2.
10. **Metric internal consistency:** $\text{tp}_t \leq n_t$ and $\text{tp}_t \leq S$ at every threshold; n_t monotone decreasing in t ; $\text{AP} \in [0, 1]$; $|CS_{95}| \leq p$; band counts summing to p ; causal band counts summing to S . Zero violations in every case.
11. **Negative controls, exactly:** all four BEATRICE variants identical to machine precision on both unannotated arms ($\max |\text{diff}| = 0$ over 625 cells), and `polyfun_{est,ldsc,oracle}` identical to `susie` there.
12. **Monotonicity where it should hold:** AP strictly decreasing in S ; strictly increasing in p_{causal} ; sparse arm easier than sparse+infinitesimal. AP is *not* monotone in ϕ , which §5 shows to be a genuine method-specific effect rather than a fault.
13. **Simulation code:** PVE calibration verified to be exact by construction in both architectures; the oracle prior verified to use the identical formula to the causal-selection weights.

13 Limitations

1. **In-sample LD only.** Iteration 003 established that in-sample and reference-panel LD can invert method rankings. Nothing here transfers to the reference-panel setting.
2. **The infinitesimal arm scores the sparse component only.** Non-causal variants carry real effects that count as false discoveries. Absolute FDR is not comparable across arms; degradation between arms is.
3. **Nominal region sizes are approximate above 1000**, with the 2000 class median 1590.
4. **Range dependence.** All importance statements are relative to the ranges swept.
5. **Per-cell decisions are underpowered (§10).**
6. **Two strata lack a noise floor** and are excluded from Sense V.
7. **κ decomposition unavailable**; the replicate table lacks the region index.
8. **One tuning configuration.** BEATRICE-family hyperparameters come from a single Op-tuna study and are held fixed; competitors run at documented defaults. A separate analysis found the family’s sensitivity to those hyperparameters to be small (± 0.004 AP across a wide hand-tuned sweep) relative to the architectural contrasts reported here.

14 Full results tables

Bold rows are the BEATRICE family. *cells* counts non-missing cells.

Table 1: **Sparse / no annotations** — all 18 analysed methods, ranked by mean AP. *cells* is non-missing cells out of 125.

method	AP	FDR@.9	mass	ECE _{hi}	CS ₉₅	cov ₉₅	RES	cells
susie_inf	0.7095	0.0062	1.14	0.0288	4.2	19.864	0.0023	125
polyfun_est	0.6819	0.0061	1.73	0.0225	1.7	19.800	0.0024	125
polyfun_ldsc	0.6819	0.0061	1.73	0.0225	1.7	19.800	0.0024	125
polyfun_oracle	0.6819	0.0061	1.73	0.0225	1.7	19.800	0.0024	125
susie	0.6819	0.0061	1.73	0.0225	1.7	19.800	0.0024	125
finemap	0.6767	0.0442	1.16	0.0683	1.5	19.168	0.0021	125
finemap_inf	0.6711	0.1815	1.82	0.1619	1.7	16.864	0.0021	120
beatrice	0.6659	0.0237	1.84	0.1434	1.9	18.888	0.0018	125
fb_pooled	0.6659	0.0237	1.84	0.1434	1.9	18.888	0.0018	125
fb_xregion	0.6659	0.0237	1.84	0.1434	1.9	18.888	0.0018	125
functional_beatrice	0.6659	0.0237	1.84	0.1434	1.9	18.888	0.0018	125
sparsepro	0.5737	0.1556	1.45	0.1232	1.5	17.336	0.0019	125
finimom	0.5735	0.1213	1.12	0.0890	1.7	18.056	0.0019	125
paintor	0.5247	0.0570	0.74	0.0615	1.8	19.008	0.0013	125
sbayesrc	0.4841	0.3705	10.02	0.2207	1.0	13.064	0.0015	125
marginal_z	0.4086	0.0000	0.43	NaN	735.6	20.000	0.0000	125
abf	0.4005	0.0054	0.43	0.0237	3.3	19.920	0.0008	125
funmap	NaN	0.0000	NaN	NaN	NaN	0.000	NaN	0

Table 2: **Sparse / binary** — all 18 analysed methods, ranked by mean AP. *cells* is non-missing cells out of 500.

method	AP	FDR@.9	mass	ECE _{hi}	CS ₉₅	cov ₉₅	RES	cells
polyfun_oracle	0.8432	0.0110	1.63	0.0715	1.2	19.866	0.0032	500
polyfun_est	0.7914	0.0046	1.81	0.0376	1.4	19.914	0.0028	500
polyfun_ldsc	0.7732	0.0209	1.81	0.0440	1.3	19.668	0.0027	500
fb_xregion	0.7484	0.0569	2.08	0.1454	1.3	19.306	0.0023	500
susie_inf	0.7470	0.0043	1.16	0.0283	2.1	19.870	0.0026	500
funmap	0.7462	0.0866	1.81	0.0746	1.3	18.616	0.0026	500
functional_beatrice	0.7280	0.0783	2.10	0.1547	1.3	19.034	0.0022	500
susie	0.7232	0.0050	1.80	0.0218	1.4	19.856	0.0026	500
finemap	0.7147	0.0447	1.14	0.0648	1.3	19.346	0.0023	500
finemap_inf	0.7096	0.1717	1.76	0.1585	1.5	16.894	0.0024	480
beatrice	0.7026	0.0235	1.79	0.1592	1.9	19.050	0.0021	500
fb_pooled	0.6661	0.0745	2.09	0.1547	1.5	18.312	0.0019	500
paintor	0.6458	0.0668	0.74	0.0919	1.1	19.064	0.0016	500
finimom	0.6182	0.1156	1.09	0.0876	1.4	18.264	0.0022	500
sparsepro	0.6107	0.1509	1.45	0.1232	1.3	17.504	0.0021	500
sbayesrc	0.5086	0.3592	4.74	0.2470	1.0	13.186	0.0018	500
marginal_z	0.4467	0.0000	0.43	NaN	765.5	20.000	0.0000	500
abf	0.4357	0.0047	0.43	0.0255	2.7	19.914	0.0009	500

Table 3: **Sparse / continuous** — all 18 analysed methods, ranked by mean AP. *cells* is non-missing cells out of 500.

method	AP	FDR@.9	mass	ECE _{hi}	CS ₉₅	cov ₉₅	RES	cells
polyfun_oracle	0.9114	0.0119	1.21	0.0791	1.0	19.952	0.0037	500
polyfun_est	0.8061	0.0051	1.77	0.0414	1.6	19.928	0.0028	500
fb_xregion	0.7933	0.0491	2.14	0.1529	1.3	19.462	0.0027	500
polyfun_ldsc	0.7914	0.0047	1.77	0.0360	1.6	19.906	0.0028	500
fb_pooled	0.7681	0.0422	2.15	0.1504	1.6	19.424	0.0023	500
funmap	0.7617	0.0227	1.77	0.0367	1.6	19.610	0.0027	500
susie_inf	0.7398	0.0053	1.13	0.0284	3.0	19.886	0.0026	500
functional_beatrice	0.7204	0.1002	2.27	0.1728	1.2	18.680	0.0022	500
susie	0.7175	0.0059	1.77	0.0236	1.6	19.852	0.0027	500
finemap	0.7071	0.0492	1.16	0.0682	1.4	19.250	0.0023	500
finemap_inf	0.7053	0.1708	1.77	0.1555	1.6	16.870	0.0024	480
paintor	0.7001	0.0520	0.74	0.0790	1.0	19.420	0.0019	500
beatrice	0.6981	0.0198	1.82	0.1531	1.9	18.846	0.0021	500
sbayesrc	0.6192	0.3000	5.75	0.1993	1.0	14.482	0.0024	500
finimom	0.6177	0.1152	1.09	0.0910	1.6	18.308	0.0022	500
sparsepro	0.6093	0.1502	1.45	0.1236	1.4	17.636	0.0021	500
marginal_z	0.4371	0.0000	0.43	NaN	738.4	20.000	0.0000	500
abf	0.4271	0.0024	0.43	0.0237	3.0	19.910	0.0009	500

Table 4: **Sparse+infinitesimal / no annotations** — all 18 analysed methods, ranked by mean AP. *cells* is non-missing cells out of 500.

method	AP	FDR@.9	mass	ECE _{hi}	CS ₉₅	cov ₉₅	RES	cells
finemap_inf	0.5945	0.1229	1.44	0.1331	62.4	18.810	0.0015	500
susie_inf	0.5939	0.0399	2.89	0.0742	42.2	19.350	0.0016	500
beatrice	0.5696	0.0087	2.52	0.1263	2.5	18.132	0.0014	500
fb_pooled	0.5696	0.0087	2.52	0.1263	2.5	18.132	0.0014	500
fb_xregion	0.5696	0.0087	2.52	0.1263	2.5	18.132	0.0014	500
functional_beatrice	0.5696	0.0087	2.52	0.1263	2.5	18.132	0.0014	500
polyfun_est	0.5208	0.3475	4.22	0.2162	1.3	16.696	0.0012	500
polyfun_ldsc	0.5208	0.3475	4.22	0.2162	1.3	16.696	0.0012	500
polyfun_oracle	0.5208	0.3475	4.22	0.2162	1.3	16.696	0.0012	500
susie	0.5208	0.3475	4.22	0.2162	1.3	16.696	0.0012	500
finemap	0.5137	0.2519	1.97	0.1920	1.3	16.380	0.0012	500
sparsepro	0.4753	0.2911	2.01	0.2076	1.4	15.836	0.0012	500
paintor	0.4386	0.1289	0.85	0.1213	1.8	16.796	0.0010	500
finimom	0.4208	0.3268	2.34	0.2131	1.5	14.262	0.0011	500
sbayesrc	0.3837	0.4968	16.33	0.2317	1.1	11.610	0.0010	500
marginal_z	0.3606	0.0000	0.43	NaN	765.9	19.990	0.0000	500
abf	0.3583	0.0903	0.43	0.0772	4.2	17.118	0.0007	500
funmap	NaN	0.0000	NaN	NaN	NaN	0.000	NaN	0

Table 5: **Sparse+infinitesimal / binary** — all 18 analysed methods, ranked by mean AP. *cells* is non-missing cells out of 2,000.

method	AP	FDR@.9	mass	ECE _{hi}	CS ₉₅	cov ₉₅	RES	cells
polyfun_oracle	0.6310	0.3565	4.05	0.2666	1.1	17.737	0.0015	2000
finemap_inf	0.6060	0.1152	1.42	0.1305	64.0	18.889	0.0016	2000
susie_inf	0.6034	0.0373	2.87	0.0719	44.2	19.400	0.0017	2000
fb_xregion	0.6034	0.0793	3.37	0.1680	1.7	18.114	0.0015	2000
functional_beatrice	0.5938	0.1167	3.45	0.1893	1.5	17.915	0.0015	2000
beatrice	0.5789	0.0094	2.51	0.1303	2.6	18.222	0.0014	2000
polyfun_est	0.5480	0.3462	4.22	0.2162	1.3	16.873	0.0012	2000
polyfun_ldsc	0.5443	0.3786	4.21	0.2379	1.3	16.818	0.0011	2000
susie	0.5304	0.3470	4.22	0.2168	1.3	16.737	0.0012	2000
fb_pooled	0.5296	0.1029	3.27	0.1726	1.7	16.663	0.0012	2000
finemap	0.5221	0.2487	1.97	0.1935	1.3	16.525	0.0012	2000
paintor	0.5076	0.1664	0.85	0.1705	1.3	17.077	0.0011	2000
sparsepro	0.4831	0.2906	2.00	0.2097	1.4	15.902	0.0012	2000
funmap	0.4551	0.5143	4.22	0.2968	1.1	13.431	0.0011	2000
finimom	0.4305	0.3215	2.36	0.2131	1.4	14.386	0.0011	2000
sbayesrc	0.3812	0.5143	15.68	0.2326	1.1	11.367	0.0010	2000
marginal_z	0.3679	0.0000	0.43	NaN	786.4	19.991	0.0000	2000
abf	0.3661	0.0910	0.43	0.0795	4.2	17.074	0.0007	2000

Table 6: **Sparse+infinitesimal / continuous** — all 18 analysed methods, ranked by mean AP. *cells* is non-missing cells out of 2,000.

method	AP	FDR@.9	mass	ECE _{hi}	CS ₉₅	cov ₉₅	RES	cells
polyfun_oracle	0.7526	0.3834	3.21	0.3195	1.1	18.925	0.0019	2000
fb_xregion	0.6521	0.0675	3.37	0.1697	1.6	18.706	0.0018	2000
fb_pooled	0.6205	0.0519	3.11	0.1593	1.7	18.260	0.0015	2000
finemap_inf	0.6135	0.1163	1.41	0.1312	61.0	18.897	0.0016	2000
susie_inf	0.6120	0.0368	2.83	0.0741	42.5	19.434	0.0017	2000
functional_beatrice	0.5999	0.1573	3.59	0.2128	1.4	17.820	0.0015	2000
beatrice	0.5883	0.0082	2.51	0.1334	2.5	18.376	0.0015	2000
paintor	0.5687	0.1544	0.84	0.1770	1.1	18.044	0.0013	2000
polyfun_ldsc	0.5604	0.3510	4.22	0.2191	1.3	17.038	0.0013	2000
polyfun_est	0.5603	0.3518	4.22	0.2190	1.3	17.031	0.0013	2000
susie	0.5383	0.3534	4.22	0.2198	1.3	16.852	0.0012	2000
finemap	0.5289	0.2550	1.97	0.1959	1.3	16.628	0.0013	2000
sparsepro	0.4883	0.2980	2.00	0.2133	1.4	15.985	0.0012	2000
sbayesrc	0.4636	0.4554	18.43	0.2112	1.0	12.875	0.0013	2000
funmap	0.4611	0.4138	4.22	0.2371	1.2	13.826	0.0012	2000
finimom	0.4352	0.3276	2.35	0.2164	1.4	14.426	0.0011	2000
marginal_z	0.3743	0.0000	0.43	NaN	765.3	19.992	0.0000	2000
abf	0.3724	0.0948	0.43	0.0774	4.1	17.174	0.0007	2000

Table 7: Paired ablation on AP, all six strata. Each row compares two methods on the *same* simulated replicate.

stratum	contrast	$\bar{d}$	sd	n	t
Sparse / no annotations	annotations	+0.0000	0.0000	1,250	—
Sparse / no annotations	region sharing	+0.0000	0.0000	1,250	—
Sparse / no annotations	combined	+0.0000	0.0000	1,250	—
Sparse / binary	annotations	+0.0254	0.0975	4,981	18.4
Sparse / binary	region sharing	+0.0205	0.0859	4,980	16.8
Sparse / binary	combined	+0.0458	0.1026	4,980	31.5
Sparse / continuous	annotations	+0.0224	0.0962	4,965	16.4
Sparse / continuous	region sharing	+0.0730	0.1183	4,960	43.5
Sparse / continuous	combined	+0.0954	0.1352	4,960	49.7
Sparse+infinitesimal / no annotations	annotations	+0.0000	0.0000	5,000	—
Sparse+infinitesimal / no annotations	region sharing	+0.0000	0.0000	5,000	—
Sparse+infinitesimal / no annotations	combined	+0.0000	0.0000	5,000	—
Sparse+infinitesimal / binary	annotations	+0.0149	0.0905	20,000	23.3
Sparse+infinitesimal / binary	region sharing	+0.0096	0.0797	20,000	17.0
Sparse+infinitesimal / binary	combined	+0.0244	0.0938	20,000	36.9
Sparse+infinitesimal / continuous	annotations	+0.0116	0.0909	20,000	18.0
Sparse+infinitesimal / continuous	region sharing	+0.0522	0.0989	20,000	74.6
Sparse+infinitesimal / continuous	combined	+0.0638	0.1102	20,000	81.9